

복사전달 특강

Special Topics in Radiative Transfer

Lecture 9
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Based on: Chapter 7 of Noebauer & Sim (Living Reviews in Computational Astrophysics)

선광일 (Kwangil Seon)
KASI / UST

Chapter 7: Thermal and Line Emission in MCRT

Overview of approaches to emission in MCRT

- Absorption/scattering treatment in MCRT: standard and well-established
- Emission handling: varied, complex, and site of ongoing developments

Focus: emissivity **within** the computational domain (not boundary injection, cf. Sect. 5.3)

7.1 Known Emissivity

- Emissivity given externally
- Direct sampling — no self-consistency
- Example: non-radiative heating

7.2 Radiative Equilibrium (RE)

- Emissivity set by radiation field itself
- Local balance: absorption = emission
- Cannot be known without RT simulation

7.3 RE by Iteration

- Iterative RT simulations
- Energy conservation only asymptotic
- Convergence issues at high optical depth

7.4 On-the-fly RE (Indivisible Packets)

- MC quanta never destroyed or degraded
- Strict RE at every packet event
- Macro atom method (Lucy 2002, 2003)

7.5 Beyond Radiative Equilibrium

External non-radiative heating (H_E) and expansion cooling (C_E) added as source/sink terms
Applied in: stellar outflows, SNe codes, accretion disc winds

7.1 Known Emissivity

Simplest case: emissivity given without prior knowledge of the radiation field

- Applies when emissivity is externally known or easily estimated independently
- Example: non-equilibrium plasma heated/ionised by non-radiative processes

Procedure

1. Sample emissivity using standard sampling techniques (Sect. 4.2)
2. Create packets representing the emission process:
 - Photon frequency ν — sampled from the emissivity spectrum
 - Weight / energy — set by the emissivity normalisation
 - Propagation direction — sampled from the angular distribution
3. Inject alongside any packets from external boundary conditions
4. Follow their subsequent propagation in the same way

Time-dependent Extension

In time-dependent simulations, emission packets are injected gradually over the time step. The injection time t_{inj} is an additional property to be sampled (uniformly over Δt).

Key advantage: no iteration required — emissivity is decoupled from the radiation field. Simply add these source packets alongside any other packet populations and propagate.

7.2 Radiative Equilibrium (RE)

When the emissivity is determined by the radiation field itself

What is Radiative Equilibrium?

At every point in the medium there is local balance between absorption and emission of radiation:

$$\text{Rate of absorption} = \text{Rate of emission}$$

Emissivity is effectively determined by the radiation field (near-equilibrium) and cannot be anticipated independently of a RT simulation.

Astrophysical Applications

- Stellar and disk atmospheres
- Hot-star winds
- Type Ia Supernovae (SN Ia)
- Cataclysmic variable accretion disc winds
- Active galactic nuclei winds

→ RE is a good approximation for many key RT problems in astrophysics

****Challenge:**** Emissivity depends on the radiation field, but the radiation field depends on the emissivity → must be solved self-consistently.

- Two strategies:
- Sect. 7.3 — RE by iteration: run RT repeatedly until convergence
- Sect. 7.4 — On-the-fly: use indivisible energy packets so RE is enforced at every event

7.3 Radiative Equilibrium in MCRT by Iteration

Solving RE via a sequence of radiative transfer simulations

Iterative Scheme

1. Start with initial estimate of medium conditions (T, ionisation state, level populations...)
2. Calculate emissivity from current conditions
3. Run MCRT simulation → extract updated radiation field estimators
4. Use RT outcomes to obtain improved estimate of conditions
5. Repeat steps 2–4 until convergence

When It Works

- Applicable to photon-packet and energy-packet schemes
- Used for part of emissivity, e.g. radiative cooling (Long & Knigge 2002)
- Works when optical depth is not too severe

Convergence Problems

- Non-local RT character → slow convergence
- High optical depth: especially severe
- Energy conservation only asymptotic
- Over/under-estimated emissivities create spurious energy sources and sinks

Historical Perspective

Convergence difficulties are well known from stellar atmosphere modelling (Hubeny & Mihalas 2014). The non-local character of RT inhibits convergence at high optical depth in the pure iterative form.

Motivation for Sect. 7.4: enforce RE exactly at every packet event without any iteration.

7.4 On-the-fly RE via Indivisible Energy Packets

The key idea: enforce energy conservation as a Monte Carlo rule

- Developed by Lucy (Abbott & Lucy 1985; Lucy 1999a, 2005)

****Key Principle:**** RE requires local balance between absorption and emission of energy.
In an energy-packet discretisation, MC quanta already represent bundles of radiative energy.

→ MC quanta are NEVER destroyed or degraded in weight during the simulation.
→ ALL interactions — even those representing pure absorption — become effective scatterings.

Indivisible Packet Principle

- Each MC packet carries fixed energy ϵ (invariant)
- Absorption event → packet immediately re-emitted (in situ)
- Packet energy conserved at every interaction
- Strict RE enforced — no spurious sources/sinks

Contrast with Sect. 7.3: RE enforced locally at each event, not asymptotically at convergence

Rules for Effective Scattering

The rules for how a packet's properties (frequency, direction) change depend on the physical process:

- Statistical equilibrium → traffic flow rules (Sect. 7.4.1–7.4.4)
- Thermal equilibrium (TE) → k-packet rules (Sect. 7.4.5)

Formulated generally by Lucy (2002, 2003)

7.4.1 Example: Effective Resonant Scattering in a Two-level Atom

Simplest illustration of the indivisible packet RE principle (Lucy 1999a)

- Spectral line absorption in a two-level atom (see Fig. 5 →)
- Radiation dominates excitation/de-excitation; isotropic emission
- Stimulated emission treated as negative absorption

Line Emissivity — Eq. (57)

$$\eta_{ul} = (h\nu / 4\pi) \cdot \psi_{ul}(\nu) \cdot R_{ul}$$

$\psi_{ul}(\nu)$ = emission profile function (normalised: $\int \psi_{ul} d\nu = 1$)

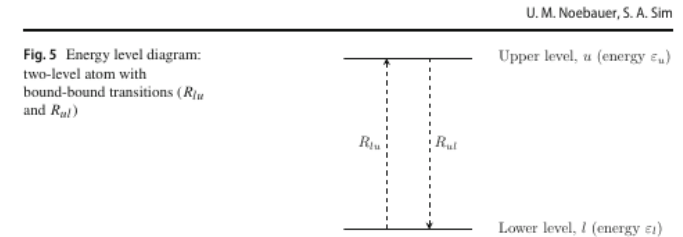
Transition Rate — Eq. (58)

$$R_{ul} = n_u \cdot A_{ul}$$

n_u = upper-level population, A_{ul} = Einstein-A coefficient

The Problem

Need n_u to evaluate η_{ul} , but n_u is unknown until after the RT simulation.
→ Statistical equilibrium provides the resolution (see next slide).



we will elaborate these principles more generally, but for concreteness we first discuss

7.4.1 MCRT Rule for Two-level Atom

Statistical equilibrium as a traffic flow condition

****Statistical Equilibrium — Eq. (59):**** $R_{lu} = R_{ul}$

Rate of absorption from the radiation field = Rate of emission into the radiation field

MCRT Implementation — Traffic Flow Rule

Whenever an MC energy packet is absorbed by the spectral line → it is immediately (in situ) re-radiated by the same line.

- Absorption of packet $\leftrightarrow$ realisation of R_{lu} (inflow term)
- Re-emission of packet $\leftrightarrow$ realisation of R_{ul} (outflow term)

New direction and frequency sampled from $\psi_{ul}(v)$; packet energy ϵ conserved.
Eq. (59) is thus embedded as a processing rule — no knowledge of n_u needed.

Key Features

- Leading n_u dependence cancels via normalisation
- Strict energy conservation at every event
- MC packets never destroyed
- Simple and fast to implement
- Used in: Abbott & Lucy 1985; Long & Knigge 2002; Kusterer et al. 2014

Physical Justification & Limitations

- Well justified for strong UV metal lines:
C IV 1550 Å, N V 1242 Å in stellar winds
- Limitation: two-level approximation
— not realistic when fluorescence or thermal reprocessing is important
→ Extended in Sects. 7.4.2–7.4.4

7.4.2 Fluorescence & Thermal Emissivity via Redistribution Parameters

Extending indivisible packets to handle flux redistribution

- Redistribution parameter Λ introduced (Baron et al. 1996; Pinto & Eastman 2000a,b; Kasen et al. 2006)

At each packet interaction: draw random number $\xi \in [0, 1]$ and compare to Λ :

$\xi > \Lambda \rightarrow$ Coherent scattering: new direction assigned; CMF frequency conserved
(same as electron scattering or resonance line scattering)

$\xi \leq \Lambda \rightarrow$ Incoherent (effective) scattering: new direction + new frequency
drawn from a normalised emissivity distribution, e.g. LTE thermal: $\chi_\nu B_\nu$

Setting Λ

- Global constant, or per-interaction type
- For line scattering: estimate by comparing collisional vs. radiative de-excitation rates (similar to Long & Knigge 2002)
- $\Lambda \rightarrow 0$: pure redistribution (like Sect. 7.1 but packet energies remain fixed!)
- $\Lambda \rightarrow 1$: nearly coherent scattering

Demonstrated Success for SN Ia

Baron et al. 1996; Pinto & Eastman 2000a,b;
Kasen et al. 2006; Magee et al. 2018:

Thermal redistribution (simple Λ) captures flux redistribution in SN Ia well and agrees with more detailed treatments.

****Key distinction from Sect. 7.1:**** Although the emissivity distribution shape is used to draw new frequencies, the absolute normalisation (packet energy) remains fixed $\rightarrow$ strict RE enforced.

7.4.3 Multi-level Atoms: Downbranching Scheme

Extending traffic flow rules to handle fluorescence and cascades (Lucy 1999b)

- Problem: fluorescence couples multiple spectral lines → cannot be described by a single redistribution emissivity

Downbranching Scheme — Lucy (1999b)

When an energy packet is absorbed via transition to upper level u ,
re-emit in transition $u \rightarrow l$ with probability proportional to the energy flow rate:

$$q_{\{u \rightarrow l\}} = (R_{\{ul\}} \cdot \epsilon_{\{ul\}}) / \sum_{\{k < u\}} (R_{\{uk\}} \cdot \epsilon_{\{uk\}})$$

Use energy flow rates (not pure transition rates) because packets carry fixed energy.
Only downward transitions ($u \rightarrow l$, $l < u$) are considered.

Strengths

- Large improvement over resonance scattering for SN spectra (Lucy 1999b; Mazzali 2000)
- Performs extremely well even compared to full macro atom (Kerzendorf & Sim 2014)
- Computationally efficient

Limitations

- Only downward transitions allowed
- $1 \rightarrow 2$ transition not connected to $2 \rightarrow 3$ cascade
- Cannot handle inverse fluorescence ($1 \rightarrow 2$, $2 \rightarrow 3$ absorption → $3 \rightarrow 1$ emission)
- Formally derived assuming downward rates dominate and $\epsilon_u \gg \epsilon_l$

Motivation for Macro Atom

A complete solution must include ALL transitions (upward and downward).
→ Full macro atom method (Lucy 2002, 2003)

7.4.3 The Macro Atom Method

General framework for multi-level atoms in radiative and statistical equilibrium

Central Concept — Lucy (2002, 2003)

- View all excited levels of the matter as energy pools
- Energy flows in/out of each pool via the complete set of radiative and collisional transitions
- Equilibrium condition: energy associated with each pool is stationary
→ expressed as traffic flow equations embedded in the MCRT simulation
- Crucially: include energy flow between excitation pools (internal radiationless transitions)
not just between radiation field $\leftrightarrow$ matter — this is what downbranching misses

Three Types of Events

Activation:

- Radiation packet absorbed
- Energy enters an excitation pool ($\dot{A}$ term)

Deactivation:

- Radiation packet emitted
- Energy leaves an excitation pool ($\dot{E}$ term)

Internal transition:

- Radiationless energy transfer between pools (ϵ -R terms)

Advantages over Downbranching

- Handles ALL transitions (upward + downward)
- Captures inverse fluorescence naturally
- Both two-level and downbranching are special limiting cases:

Two-level atom = macro atom, $\epsilon_1 = 0$

Downbranching = macro atom with all internal transitions suppressed

Key Result (Lucy 2002)

Level populations n_i cancel from transition probabilities → macro atom works without converged level populations. Demonstrated for Fe II (Lucy 2002, Fig. 5).

7.4.4 Macro Atom for a Three-level Atom

Deriving macro atom rules from statistical equilibrium (bound-bound only, radiation-dominated)

- Bound-bound only; radiation dominates all rates; isotropic emission; no collisions
- Goal: formulate rules for packets absorbed in lines ν_{12} , ν_{13} , ν_{23}

Statistical Equilibrium Equations — Eqs. (61)–(63)

Rates into level = rates out of level for each level:

$$\text{Level 1: } R_{21} + R_{31} - R_{12} - R_{13} = 0$$

$$\text{Level 2: } R_{12} + R_{32} - R_{21} - R_{23} = 0$$

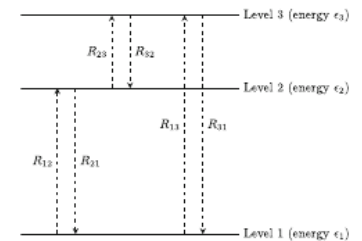
$$\text{Level 3: } R_{13} + R_{23} - R_{31} - R_{32} = 0$$

where R_{ij} = rate of transitions from level i to level j

- Multiply each equation by level energy ϵ_1 , ϵ_2 , ϵ_3 and rearrange
- $\rightarrow$ obtain energy flow budget equations for each level's excitation pool

U. M. Noebauer, S. A. Sim

Fig. 6 Energy level diagram for the simple three-level atom used as an example for the formulation of macro atom rules



state). Specifically, we identify

7.4.4 Energy Flow Rates and Macro Atom Equations

Translating statistical equilibrium into energy budgets for each level

Absorption Rates (radiation → excitation) — Eq. (64)

Energy flowing from radiation field into each pool:

$$\dot{A}_{12} = \epsilon_{12} R_{12}$$

$$\dot{A}_{13} = \epsilon_{13} R_{13}$$

$$\dot{A}_{23} = \epsilon_{23} R_{23}$$

$(\epsilon_{ij} = |\epsilon_i - \epsilon_j| = \text{energy of the transition})$

Emission Rates (excitation → radiation) — Eq. (65)

Energy flowing from each pool back to radiation:

$$\dot{E}_{21} = \epsilon_{12} R_{21}$$

$$\dot{E}_{31} = \epsilon_{13} R_{31}$$

$$\dot{E}_{32} = \epsilon_{23} R_{32}$$

Energy Pool Balance Equations — Eqs. (66)–(68) [stat. eq. × level energies, rearranged]

Level 1 ($\epsilon_1 = 0$ by convention): $\epsilon_1 R_{21} + \epsilon_1 R_{31} = \epsilon_1 R_{12} + \epsilon_1 R_{13}$ (trivially satisfied)

Level 2: $\dot{A}_{12} + \epsilon_1 R_{12} + \epsilon_2 R_{32} = \dot{E}_{21} + \epsilon_2 R_{23} + \epsilon_1 R_{21}$

Level 3: $\dot{A}_{13} + \dot{A}_{23} + \epsilon_1 R_{13} + \epsilon_2 R_{23} = \dot{E}_{31} + \dot{E}_{32} + \epsilon_2 R_{32} + \epsilon_1 R_{31}$

$\dot{A}$ terms (LHS): radiation absorbed → realised as packet absorption events

$\dot{E}$ terms (RHS): radiation emitted → realised as packet emission events

$\epsilon \cdot R$ terms: internal transitions — appear as source in one equation, sink in another

Internal $\epsilon \cdot R$ terms conserve energy by routing it between excitation pools without involving the radiation field — the essence of the macro atom.

7.4.4 Macro Atom Algorithm

Traffic flow interpretation → Monte Carlo processing rules

(A) Activation

Whenever an active radiation packet is absorbed by any of the spectral lines,
→ interpret as realisation of the corresponding $\dot{A}$ absorption term
→ The macro atom is said to be activated in the corresponding energy level

(B) Select Outcome

Inspect all sink terms on the RHS for the activated level; draw a random number ξ .
Select outcome with probability proportional to the energy flow rate. Example (level 2):
prob. $\dot{E}_{21} / D_2 \rightarrow$ emit in $2 \rightarrow 1$ transition (deactivation)
prob. $\epsilon_2 R_{23} / D_2 \rightarrow$ internal transition $2 \rightarrow 3$ (stays active)
prob. $\epsilon_1 R_{21} / D_2 \rightarrow$ internal transition $2 \rightarrow 1$ (stays active)
where $D_2 = \dot{E}_{21} + \epsilon_2 R_{23} + \epsilon_1 R_{21}$ (normalisation factor)

(C-i) Deactivation

If an $\dot{E}$ (emission) term selected:
→ macro atom deactivates
→ packet returned to main MC simulation
with new frequency and direction
(sampled from the emission process)
→ packet energy ϵ conserved (RE enforced)

(C-ii) Internal Transition

If an internal $\epsilon \cdot R$ term selected:
→ macro atom remains active, switches state
(e.g. selecting $\epsilon_2 R_{23} / D_2 \rightarrow$ state $2 \rightarrow 3$)
→ return to step (B) and repeat
→ continue until a deactivation event
($\dot{E}$ term) is drawn

7.4.4 Key Properties of the Macro Atom Scheme

Important features, special cases, and computational considerations

Level Population Cancellation

All rates $R_{ij} \propto n_i$ (level population)
→ leading n_i dependence cancels in step (B) normalisation

→ Macro atom transition probabilities can be computed without converged level populations!

Demonstrated for Fe II (Lucy 2002, Fig. 5)

Ground State Convention

$\epsilon_1 = 0$ (standard zero of excitation energy)
→ No $\dot{A}$ or $\dot{E}$ terms for level 1
→ Internal $\epsilon_1 \cdot R$ terms vanish

→ Simplifies algorithm: ground state excluded from explicit treatment

Special Cases — Both are limits of the full macro atom

- Effective resonance scattering (Sect. 7.4.1): macro atom with only two levels, $\epsilon_1 = 0$
- Downbranching scheme (Lucy 1999b, Sect. 7.4.3): macro atom with all internal $\epsilon \cdot R$ transition terms suppressed (valid when downward rates dominate, $\epsilon_u \gg \epsilon_l$)

Computational Efficiency

Repeated cycling (B)→(C-ii) can be costly for large atomic models.

Ergon et al. (2018): Markov-chain approach solves the macro atom without explicitly following internal state transitions → substantial reduction in computational cost.

- Note: Additional population dependence can re-enter via stimulated emission corrections and Sobolev escape probabilities (Sect. 8.2)

7.4.5 The Thermal Energy Pool (k-packets)

Extending the macro atom to include particle kinetic energy (Lucy 2002)

Generalise the macro atom to include the **thermal pool** of particle kinetic energies

Thermal Equilibrium (TE) Condition — Eq. (69):

$$H_R + H_C = C_R + C_C$$

H_R = radiative heating (e.g. free-free absorption) C_R = radiative cooling

H_C = collisional heating (de-excitation of atoms) C_C = collisional cooling (excitation)

k-packet Processing Algorithm

When a radiation packet is absorbed via a heating process (realising H_R or H_C):

→ Energy flows into the thermal pool (a 'k-packet' event)

→ Sample all available cooling processes with probability proportional to their rates:

Prob. $C_R / (C_R + C_C)$: re-emit a radiation packet
(frequency and direction from cooling emissivity)

Prob. $C_C / (C_R + C_C)$: activate a macro atom at the collisionally excited level
(coupling thermal pool → excitation pool)

Cross-talk Between Pools

- Bound-free (photoionization/recombination) processes involve BOTH atomic and thermal pools simultaneously; handled by extending the traffic flow equations (Lucy 2003)
- Energy can cycle: radiation ↔ excitation pool ↔ thermal pool — all conserved exactly

7.5 Indivisible Energy Packets Beyond Radiative Equilibrium

Generalising to include external heating and non-equilibrium sinks

- When RE does not hold: add external source/sink terms to the energy balance equation

Generalised Energy Balance — Eq. (70) [e.g. stellar outflow]:

$$H_R + H_C + H_E = C_R + C_C + C_E$$

H_E = external non-radiative heating (e.g. MHD heating) C_E = external cooling (adiabatic expansion)

Treating H_E (External Heating)

- H_E packets injected directly into the k-packet pool during the simulation
- Injection rate set by the external non-radiative process
- Energy initially enters as thermal kinetic energy
- Subsequently processed via the standard TE traffic flow rules (C_R , C_C channels)

Treating C_E (External Cooling / Sinks)

- Option 1 — Packet Termination:
Prob. $C_E / (C_R + C_C + C_E)$:
packet terminated (energy removed)
- Option 2 — Energy Reduction:
Sample only C_R , C_C channels;
reduce packet energy by a factor
accounting for expansion loss

Applications & Limiting Behaviour

- Adiabatic cooling implemented in: Long & Knigge 2002; Kasen et al. 2006; Kromer 2009; Vogl et al. 2019
- In the limit H_E , C_E dominant → reduces to the known-emissivity scheme of Sect. 7.1
- Extensions to non-statistical-equilibrium or non-TE terms not yet implemented (future work)

Summary: Chapter 7 — Emission Treatments in MCRT

Hierarchy from simple to general

7.1 Known Emissivity

- Emissivity given externally
- Direct sampling — no iteration
- Simple injection alongside other packets

7.2–7.3 RE by Iteration

- Iterative RT cycles
- Energy conservation only asymptotic
- Convergence issues at high τ

7.4 On-the-fly RE

- Indivisible energy packets
- Strict RE at every interaction
- Lucy 1999a, 2002, 2003

7.4.1–7.4.2

- Two-level: resonant scattering
- Redistribution parameter Λ
- Simple, fast; limited realism

7.4.3 Downbranching

- $q_{\{u \rightarrow l\}} \propto$ energy flow rates
- Multi-level downward only
- Excellent for SN Ia spectra

7.4.3–7.4.4 Macro Atom

- All transitions (up + down)
- Level pops cancel from rates
- Most complete treatment

7.4.5 k-packets

- Thermal energy pool added
- $H_R + H_C = C_R + C_C$ (TE)
- Couples to macro atom

7.5 Beyond RE

- External H_E , C_E terms
- Packet termination / reduction
- Adiabatic cooling in SNe codes

Key Achievement

- Level pops cancel $\rightarrow$ no need for converged populations
- Macro atom = general framework